

CHAINLAB

Solana Fundamentals

Accounts, transactions, programs and network concepts

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Describe Solana's account-oriented model.
- Explain programs, instructions, transactions and fees.
- Understand clusters and basic development workflow.

1. Solana overview

- Solana is a blockchain network designed for high-throughput applications. Its runtime, transaction format, validator architecture and fee model differ from those of other chains.
- Performance claims should be evaluated with workload, hardware, network conditions and measurement methods in mind.

2. Accounts and programs

- On Solana, state is stored in accounts. Accounts have an address, owner, data, lamports and metadata such as executable status.
- Programs contain executable logic. A program generally receives account references and instruction data, then validates and updates permitted account state.
- Program-derived addresses (PDAs) are deterministic addresses derived from seeds and a program ID; they are not ordinary keypairs with a private key.

3. Instructions and transactions

- An instruction specifies a program, accounts and instruction data. A transaction can contain multiple instructions and is signed by the required signers.
- Transactions must satisfy runtime rules, account access requirements, compute limits and fee requirements. They can fail, and clients should handle errors safely.

4. Clusters and development

- Common environments include local development, Devnet for testing, Testnet for network testing and Mainnet Beta for production use.
- Use test tokens and test environments while learning. Confirm the selected cluster before signing or sending transactions.

5. Fees and safety

- Fees compensate network processing and may include a base fee and optional prioritization components, depending on transaction conditions.
- Never expose a seed phrase or private key in source code, browser storage, screenshots or public repositories.

Review questions

- What does an account's owner field indicate?
- How is a PDA different from a normal keypair address?
- Why should developers test on Devnet before production deployment?

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